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# Remarkably Improved Cycling Stability of Boron-strengthened Multicomponent Layer Protected Micron-Si Composite Anode

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**ABSTRACT:** The development of micron-sized Si as an advanced anode for Li-ion batteries remains a huge challenge in alleviating its vast volume expansion during lithiation. Herein, we successfully prepare a spherical Si composite coated by a boron–strengthened multicomponent surface layer, via heating of Si particles with $LiBH_4$ followed by a mechanochemical reaction with $CO_2$. The thin high-strength outer layer consists of amorphous $SiO_x$/C, in which ultrafine nanocrystals of SiC and $Li_2SiO_3$, and amorphous B and $B_2O_3$ species, are well-dispersed. These multicomponent nanocrystals and B species reinforce effectively the mechanical properties of amorphous $SiO_x$/C conformal layer, which restrains greatly the volume expansion in the lithiation process and stabilizes the forming solid electrolyte interphase layer due to the reduced cracks and pulverization of the anode material, and consequently leads to a remarkable long-term cyclability. An initial capacity of 1385 mAh $g^{-1}$ at 100 mA $g^{-1}$ is measured with a good cycling stability which still retained 70.5% of capacity over 600 deep cycles.

**KEYWORDS**: Li-ion batteries, anode materials, silicon, complex hydrides, cycling stability

## INTRODUCTION

Li-ion batteries (LIBs) have been regarded as one of preferable power supplies for portable devices and electric vehicles (EVs) due to their high energy density and design flexibility.[1-5] To meet the ever-increasing energy storage needs for various technological applications, seeking and developing new electrode materials are of critical importance.[6-9] Si, with an extremely high theoretical capacity of 4200 mAh $g^{-1}$ (corresponding to $Li_{4.4}Si$), low working potential (0.5 V vs $Li^+/Li$), natural abundance (27.72% on earth) and environmental friendliness, has been paid extensive attention as one of the most promising anode substitutes for the commercial graphite (372 mAh $g^{-1}$).[10-12] However, the Si-based anode materials usually suffer from a fast capacity fading due to the huge volume change upon lithiation/delithiation.[13,14] Various approaches have been proposed to address this issue, which include limiting the degree of lithiation,[15,16] fabricating nanostructures and hollow structures,[17-20] compositing with inactive matrixes,[21,22] alloying with other active materials,[23-25] engineering surfaces and interfaces,[26,27] adding electrolyte additives,[28-30] and using novel binders.[31,32]

Designing and fabricating nanostructured Si is the most widely adopted strategy to improve the performance of Si anode since Cui's group demonstrated that Si nanowires can achieve its theoretical charge capacity and accommodate large strain without pulverization.[33] A variety of morphologies in nanoscale, such as nanospheres, nanotubes, nanosheets and nanocrystals, have been constructed and evaluated for their anode performance.[17-20] Nevertheless, nanostructured Si anode brings along a variety of challenges, including low packing density, uncontrolled side-reaction, and especially complicated and expensive synthesis process, which adversely makes it difficult to commercialize.[34,35] In this context, the utilization of Si particles in micron scale should be more feasible due to their widespread availability and low cost.[36-46]

For micron-sized Si anode, a number of research groups have achieved enhanced anti-pulverization performances in the lithiation/delithiation process by creating porous structure, compositing with carbon and surface engineering.[37] A large mesoporous silicon sponge (> 20 μm) prepared by electrochemical etching reduced the volumetric expansion at fully lithiated state to 30% and effectively restrained the pulverization of Si particles. As a result, a capacity of ~750 mAh $g^{-1}$ was obtained with >80% capacity retention over 1000 cycles.[38] Similarly, porous Si-C composites of micron size showed high capacity and good cycling stability.[39-41] In particular, 2-10 μm-sized Si/C composites maintained 86.8% of initial capacity (1182 mAh $g^{-1}$) after 300 cycles while operating at 50 mA $g^{-1}$.[42] Alternatively, functional coating enables a long-term cycling performance because it can buffer the volume change upon cycling and stabilizes a solid electrolyte interphase (SEI) layer as well. The frequently used coating materials are oxides, such as $SiO_x$, $TiO_2$, $Al_2O_3$, ZnO and $Li_2TiO_3$.[43] As demonstrated by Park et al.,[44] the amorphous $SiO_x$-coated Si composite provided 1914 mAh $g^{-1}$ of specific capacity with good cycling stability within 100 cycles. Lee et al. obtained 90% of capacity retention after 50 cycles at 0.1 C of rate for a Si composite coated with a multicomponent layer, which was composed of $Li_2SiO_3$, $Li_2Si_2O_5$ and $Li_4Ti_5O_{12}$.[45] Similar beneficial effect on the cycling durability was also observed by Yang and co-workers for the Si micro-particles with a nanocrystalline SiC and $Li_2SiO_3$ strengthened SiOC layer coating, which showed a capacity retention of ~ 75% after cycling 400 times at 0.1 C.[46] Apparently, multicomponent coating is a viable strategy to limit and accommodate the volume change upon cycling and modify the surface chemistry for a stable SEI protection layer.

As predicted theoretically,[47] a thin surface coating layer with stiffness higher than $Li_{15}Si_4$, the fully lithiated phase of Si at room temperature, would provide a good volume suppression effect. In this context, boron (B) should be a promising choice because of its extreme hardness (about 30-34 GPa).[48] Moreover, it is expected that the presence of B on the particle surface would favour creating a stable SEI layer since B is generally recognized as a poor electrical conductor at room temperature. We therefore tried to fabricate a novel high-strength B-containing multicomponent layer for coating Si microparticle composite, using the complex hydride $LiBH_4$ as a precursor. By ball milling the reaction product of Si and $LiBH_4$ at a given molar ratio under $CO_2$ atmosphere, a high-strength multicomponent conformal layer was formed in situ on the surface of Si microparticles. The multicomponent layer is mainly comprised of amorphous $SiO_x/C$ as well as well-dispersed nanocrystals of SiC and $Li_2SiO_3$ and amorphous B and $B_2O_3$. These nanocrystals and B species largely enhanced the mechanical performance of amorphous $SiO_x/C$ coatings, which effectively restrict the volume expansion upon lithiation and mitigate the deformation of electrode during cycling, consequently contributing to a good long-term cyclability.

## EXPERIMENTAL SECTION

**Sample preparation.** Commercially available $LiBH_4$ (purity 95%) and 1-3 μm-sized Si powders (purity 99.9%) were obtained from Alfa Aesar (USA) and Shanghai ST-NANO (China), respectively. Mixtures of Si-$x$LiBH$_4$ with $x$ = 0, 0.1, 0.2, 0.3, 0.4 were first prepared by means of ball milling at 300 rpm for 2 h using a QM-3SP4 planetary ball mill (Nanjing). A 45:1 weight ratio was adopted for ball-to-sample. The post-milled mixture was then collected within a MBRAUN 200B glovebox (Germany, $H_2O$, $O_2$: < 0.1 ppm) and transferred into a stainless-steel reactor, which was gradually heated at 5 °C min$^{-1}$ from room temperature to 550 °C. Finally, the target samples were obtained by ball milling the heat-treatment products at 400 rpm for 4 h within high-purity $CO_2$ atmosphere (Jingong Materials, China).

**Sample characterisations.** A 40 kV × 15 mA MiniFlex 600 X-ray diffractometer (Rigaku, Japan) equipped with Cu K$_\alpha$ radiation was employed to characterize structural information. The scattering angle (2$\theta$) range of 10-80° with 0.02° increments was selected for XRD data collection. A Tensor 27 unit (Bruker, Germany) operating at the transmission mode was adopted for recording fourier transform infrared (FTIR) spectra. The testing sample pellets were prepared by first mixing powders with KBr (weight ratio: 1:100) and then cold-pressed. Raman spectra were acquired on a Via-Reflex confocal Raman microscope (Renishaw plc, UK) with 532 nm of laser excitation wavelength. Solid-state magic angle spinning (MAS) NMR analysis was conducted on a Bruker AVIII-HD 400 NMR spectrometer (Germany). A ESCALAB 250Xi X-ray photoelectron spectroscopy (XPS) system (Thermo Scientific, USA) equipped with Al K X-ray source (1486.6 eV) was used to analyse the chemical states of elements. As a reference, the adventitious C 1s peak at 284.8 eV was used to calibrate the XPS data obtained. The obtained binding energy spectra were fitted with XPSPEAK software. Observation of morphologies was conducted with a scanning electron microscope (SEM, Hitachi-S4800, Japan). Before SEM observation, the cycled electrodes was washed several times with anhydrous diethyl carbonate (DEC) to remove the remaining electrolyte on the surface. The microstructures were identified with a transmission electron microscope (TEM, Tecnai G2 F20 S-TWIN, FEI, USA). A scanning transmission electron microscope (STEM, Titan G2 80-200 Chemi, FEI, USA) was employed to acquire the annular dark field images. The acceleration voltage was set at 200 kV. The test samples were prepared by a three-step process, including dispersing powders into anhydrous ethyl alcohol by ultrasonic treatment, dropping the suspension onto carbon-coated Cu grids and drying at room temperature. The distributions of elemental Si, O, C and B were identified through an energy dispersive X-Ray spectroscopy (EDX). Nanoindentation experiments were carried out on a Nano Indenter G200 instrument with a Berkowich diamond tip. The powder samples were cold-pressed into 200-μm-thick pellets at 20 MPa of pressure, and then attached to a stainless-steel substrate. A continuous stiffness measurement mode was used. The depth was limited 1000 nm with the frequency target of 45 Hz at 24 °C. A total of 5 indents were performed on each sample. A laser particle size analyzer (BT-2003, China) was used to characterize the size distribution of particles. The powdery samples were dispersed uniformly with ultrasonic oscillation in anhydrous alcohol.

**Electrochemical measurements.** The electrochemical Li storage properties of obtained materials were evaluated by assembling them into 2025 coin-type half-cells as anodes. To fabricate the working electrodes, the slurry of active materials was prepared by mixing with acetylene black and sodium alginate in deionized water, at a weight ratio of 7:2:1. The slurry was then pasted onto a piece of Cu foil with >1 mg $cm^{-2}$ of mass loading and dried under vacuum at 120 °C for 12 h. The counter/reference electrode was metallic Li foil, the separator was Celgard 2400 membrane, and the electrolyte was 1 M $LiPF_6$ solution. The electrolyte solution was mainly comprised of ethylene carbonate, diethyl carbonate and dimethyl carbonate (EC/DEC/DMC: 1:1:1 by volume) and 1 vol% fluoroethylene carbonate (FEC) solution was used as additive. Galvanostatic charge-discharge experiments were conducted at a constant temperature (26 ± 1 °C) with a Neware battery test unit (Shenzhen, China). The working potential window was set between 0.01-2.0 V (vs. $Li^+$/Li) and a current density of 100 mA $g^{-1}$ was adopted. An Arbin potentiostat (BT-2000, USA) was used to measure cyclic voltammetry (CV) behaviours, which was operated at 0.01-2.0 V (vs. $Li^+$/Li) and 0.1 mV $s^{-1}$. A Vertex electrochemical workstation (Ivium, The Netherlands) was used to record electrochemical impedance spectra (EIS). The voltage amplitude was 5 mV and the measured frequency range was 100 kHz-10 mHz.

## RESULTS AND DISCUSSION

**Preparation and characterization of multicomponent layer-protected micron-Si.** The target composites were synthesized by heating the Si-$x$$LiBH_4$ mixtures followed by ball milling within $CO_2$ atmosphere. In the first step of preparation, two new phases, $Li_{12}Si_7$ and B, were formed after heating the as-milled mixtures of Si-$x$$LiBH_4$ to 550 °C in addition to the starting Si phase, which were detected by XRD and XPS (Figure S1, Supporting Information). Reactions between Si and $LiBH_4$ took place on the surface of Si particles since $Li_{12}Si_7$ and B were mainly observed at the surface while Si was in the core position (Figure S1b-d, Supporting Information). The $Li_{12}Si_7$ phase then disappeared after ball milling under $CO_2$ atmosphere, as revealed by the XRD results (Figure 1a), while the reflections of Si still dominated the XRD profiles. For the samples prepared with higher $LiBH_4$ contents, two new Si-containing phases, identified as SiC and $Li_2SiO_3$, emerged, albeit with relatively low intensities. Meanwhile, a slightly upraised background as well as a small bump was observed at the low diffraction angles, indicating the presence of amorphous-phase.

Furthermore, amorphous species were detected to be mainly composed of $SiO_x$, C, $B_2O_3$ and B by FTIR, Raman, NMR and XPS. FTIR spectra (Figure 1b) show three vibration peaks at 1068, 807 and 465 $cm^{-1}$, which correspond to asymmetric and symmetric stretching modes and bending mode of Si-O bonding, respectively, according to the previous report.[49] Vibrations centred at 1360 (D band) and 1580 $cm^{-1}$ (G band) were detected by Raman spectra (Figure 1c), indicating the existence of elemental C.[50] The quite broad peaks and the high $I_D/I_G$ value (1.52) suggest the low graphitization degree, which is possibly caused by the ball milling process. By means of solid-state NMR (Figure 1d) the characteristic chemical shifts of amorphous $B_2O_3$ and B at 16 and 4 ppm were detected, respectively,[51,52] which was further evidenced with XPS analyses through the high-resolution B 1s XPS spectra of $B_2O_3$ and B emerged at 192.1 and 187.2 eV,[53,54] respectively (Figure 1e). As expected, there is a gradual increase in the XPS peak intensities of B 1s with increased $LiBH_4$ content. In addition, three peaks were observed at 102.6, 100.5 and 99.1 eV in the Si 2p XPS spectra (Figure 1f), corresponding to $SiO_x$, SiC and Si, respectively.[55] It should be noted that the intensities of Si 2p XPS peak assignable to $SiO_x$ gradually increased with the $LiBH_4$ content, whereas the peak belonging to Si weakened and that of SiC first increased and then decreased. This indicates the newly formed amorphous $SiO_x$ mainly stayed on the surface and encapsulated in situ the inner Si.

Morphology observation and composition analyses were further conducted by SEM, TEM and EDS. As shown in Figure 2, SEM observation unveiled well-dispersed particle morphology for the $LiBH_4$-containing composites after ball milling within $CO_2$, in contrast to the serious aggregation among particles for pristine Si. Interestingly, the particle size of the $LiBH_4$ containing composites increased gradually from ca. 200 nm to 1.5 μm with increased $LiBH_4$ content, which was further evidenced by characterizing their size distribution (Figure S2, Supporting Information). This reveals their improved anti-pulverization performance, reasonably related to the markedly strengthened mechanical properties imparted by the surface coating. Nanoindentation measurements provided quantitative evidence because the elastic modulus and hardness were remarkably increased from ca. 8.5 ± 0.8 GPa and 0.25 ± 0.04 GPa for pristine Si to 40.3 ± 2.2 GPa and 1.77 ± 0.11 GPa for the samples prepared from Si-0.3$LiBH_4$, as shown in Figure 2g-i. These values are also higher than those of the B-free micron-Si composite prepared with the identical process (Figure S3, Supporting Information), which is consistent with our speculation that the presence of B is in favor of the strengthened mechanical performance. The typical nanoindentation morphology is displayed in Figure S4 (Supporting Information).

Figure 3 presents TEM, EDS mapping and HRTEM images for the $x$ = 0.3 composite as a representative sample. Results (Figure 3a-e) showed that the O, C and B elements were detected mainly at the surface and Si in the interior part of the sphere, pointing to a surface coating structure. An additional linear scanning (Figure 3f) indicated that Si dominated the particle interior and O was essentially associated with Si, C and B in the surface layer, agreeing with the TEM and EDS mapping results. Moreover, the EDS peak intensities decreased in order O > Si > C > B, when sampling a given position in the surface layer. Further HRTEM observation (Figure 3g and h) revealed that a 30-50 nm-thick amorphous layer covering the surface of crystalline Si particle. All these results implied that the amorphous coating layer was mainly composed of $SiO_x$ and C, which we denoted as $SiO_x$/C hereafter. Furthermore, a certain number of nanocrystals measuring 5-10 nm in size dispersed well in the amorphous surface layer, which were identified to be $Li_2SiO_3$ and SiC, based on the lattice fringes separated by 0.271 and 0.251 nm, respectively, corresponding respectively to the crystal planes of (130) for $Li_2SiO_3$ and (015) for SiC. The SAED pattern shown in Figure 3i also clearly manifested the diffraction rings of $Li_2SiO_3$ and Si. Taking into comprehensive consideration of all the above-shown results, we assume that ball milling the post-heated Si-$x$$LiBH_4$ within $CO_2$ atmosphere gave rise to the formation of a surface-coated Si composite, in which the inner core is the unreacted crystalline Si and the outer layer is comprised of multicomponent amorphous $SiO_x$/C (Figure 4a). The $Li_2SiO_3$ and SiC nanocrystals along with the amorphous B and $B_2O_3$ dispersed in the amorphous $SiO_x$/C surface layer facilitated an improvement in the mechanical properties by functioning with a dispersion-strengthening effect as evidenced by the largely enhanced elastic modulus and hardness values (Figure 2g-i). This also explains reasonably the larger particle size for the higher $LiBH_4$ content after the identical ball milling treatment as observed in Figure 2a-f. In other words, the in situ formation of the well-dispersed $Li_2SiO_3$ and SiC nanocrystals and amorphous B and $B_2O_3$ species strengthens the $SiO_x$/C layer, which makes the composite particles less liable to pulverization during ball milling, consequently resulting in the increased particle size for the samples with higher $LiBH_4$ content.

**Electrochemical properties of multicomponent layer-protected micron-Si.** The microparticles of Si composites with a high-strength multicomponent $SiO_x$/C protecting layer were fabricated to wafer electrodes as the active anode materials and assembled into half cells to measure their electrochemical performance. The results are shown in Figure 4b-e. The pristine Si anode delivered 3833 mAh $g^{-1}$ of initial discharge capacity when cycling at 0.01-1.5 V (vs. $Li/Li^+$) and 100 mA $g^{-1}$ (Figure 4b), which is slightly higher than its theoretical lithiation capacity at room temperature (3579 mAh $g^{-1}$ for $Li_{15}Si_4$), possibly due to the formation of a SEI layer as widely reported.[56,57] After compositing with $LiBH_4$ and subsequent ball milling treatment, a considerable decrease was observed in the first discharge capacity with increased $LiBH_4$ content, which were 2956, 2204, 1834 and 1508 mAh $g^{-1}$ for $x$ = 0.1, 0.2, 0.3 and 0.4, respectively. Here, the decreased discharge capacity is mainly attributed to the formation of $SiO_x$ with less Li insertion activity, and $Li_2SiO_3$, SiC, B and $B_2O_3$ as well, which are nearly inactive for lithiation.[55,58] Also, the thicker surface coating layer for the sample with high $LiBH_4$ content may be unfavorable for the capacity due to the retardation of lithiation.[59,60] In the subsequent charge cycle, the specific capacities were determined to be 3331, 2414, 1768, 1385 and 1075 mAh $g^{-1}$. As a result, the first coulombic efficiency values were calculated to be 86.9%, 81.7%, 80.2%, 75.5% and 71.3%. The lowered first coulombic efficiency is closely related to the presence of $SiO_x$, which possesses the inferior coulombic efficiency as reported previously.[61] Encouragingly, the first coulombic efficiency was remarkably increased to 91.7% for the $x = 0.3$ sample by using Li powder as a Li compensation reagent (Figure S5, Supporting Information).

Figure 4c illustrates the cycling stability of the high-strength multicomponent layer-protected micron-Si composite anode at 100 mA $g^{-1}$. For pristine Si anode, the reversible capacity was rapidly reduced to 378 mAh $g^{-1}$ after 60 cycles and to 246 mAh $g^{-1}$ after 200 cycles, the capacity retention of which was calculated to be only 7.4% (Figure 4d). However, for the micron-Si composite anode protected by a high-strength multicomponent $SiO_x$/C layer, the specific capacities maintained at 993, 984, 1080, and 843 mAh $g^{-1}$ for $x = 0.1$, 0.2, 0.3 and 0.4, respectively, under identical cycling conditions, which correspond to 41.1%, 55.6%, 78.0%, and 78.4% of capacity retention. Here, it is worth highlighting that the samples with the highest $LiBH_4$ content exhibited more than 10-fold increase in the capacity retention with respect to pristine Si anode. This is highly related to the restricted volume expansion of Si during lithiation with the presence of strong surface layer.[59,60] Taking into consideration of both the specific capacity and the cycling stability, the micron-Si composite anode prepared from the $x = 0.3$ mixture exhibited the best overall electrochemical performance in this case. The measurement of long-term cyclability (Figure 4e) indicates that the specific capacity still stayed at 975 mAh $g^{-1}$

even after 600 deep cycles. The available capacity is still 2.6 times of that of commercial graphite (372 mAh $g^{-1}$). The corresponding capacity retention was 70.4%, i.e., more than 8-fold higher than that of pristine Si. As such, a significant improvement was achieved after encapsulating a high-strength multicomponent $SiO_x$/C layer.

**Mechanism for improved long-term cyclability.** In order to elucidate the significantly improved cycling durability, the EIS data of pristine Si and the sample prepared from Si-0.3$LiBH_4$ were collected and compared after various cycles. As shown in Figure 5a and b, both samples exhibited very similar Nyquist plots as they were composed of two semicircles followed by a straight-line portion at the high- and meddle frequency region and the low frequency range, respectively. As reported extensively,[62] the semicircle at high frequency is related to the impedance of the surface SEI film ($R_{sei}$), and that at middle frequency is connected with the charge transfer resistance ($R_{ct}$) occurring at the electrode/electrolyte interface, whereas the low frequency inclined line represents Warburg impedance ($W_o$) related to the lithium-ions diffusion in the electrode. Obviously, the diameters of both semicircles became bigger and bigger upon cycling due to the increased values of $R_{sei}$ and $R_{ct}$., The $R_{sei}$ and $R_{ct}$ values were extracted by fitting the impedance data and plotted in Figure 5c and d. The results show that the $R_{sei}$ and $R_{ct}$ values of the pristine Si anode were drastically increased from 12.3/80.2 Ω to 31.6/286.7 Ω. In comparison, only a slight increase from 13.7/38.7 to 25.3/191.1 Ω was observed for the micron-Si composite anode prepared from Si-0.3$LiBH_4$, implying a relatively stable surface chemical state. This is possibly due to the mitigated pulverization and fracture brought by the enhanced mechanical property, which is supported by the STEM observation (Figure 6a and b). While severe pulverization, crack and fracture were clearly observed for the pristine Si sample only after 50 cycles, the sample prepared from Si-0.3$LiBH_4$ displayed a good integrity even after 200 cycles.

Shown in Figure 6c and d are the high-resolution F 1s XPS spectra of the pristine Si electrode and the sample from Si-0.3$LiBH_4$ after different cycles. Here, the cycled electrodes were washed with DEC to remove the electrolyte remaining on the surface before XPS measurements. A quite asymmetric F 1s XPS peak was detected for both samples after the first cycle, which can be resolved into two peaks at around 686.8 and 684.9 eV, corresponding to F in LiF and $Li_xPF_y$,[63] respectively. Cycling induced increase in the intensities of F 1s XPS peak assignable to LiF and decrease of F 1s XPS peak for $Li_xPF_y$, indicating the gradual formation of a SEI layer on the particle surface because LiF was indicated to be one of the important SEI components in the $LiPF_6$-based electrolyte. Encouragingly, a constant XPS spectrum was reached for the Si-0.3$LiBH_4$ sample only after 5 cycles, representing a stable SEI layer upon cycling. In contrast, the relative intensities of LiF and $Li_xPF_y$ changed continuously for pristine Si electrode, implying an unstable SEI layer. In addition, STEM results shown in the right side of Figure 6a and b provided sound evidence as the multicomponent-strengthened micro-Si composite exhibited a relatively integrated and uniform SEI layer with a thickness of ~ 42 nm, while the SEI layer was not uniform and even broken for pristine Si particle. This fact reveals unambiguously that the SiC and $Li_2SiO_3$ nanocrystals and amorphous B and $B_2O_3$-containing $SiO_x$/C layer provide certain kind of protection against the pulverization and fracture of Si composite particles by limiting the volume expansion upon lithiation, consequently stabilizing the SEI layer formed in the initial cycling stage. However, for the pristine Si, the SEI layer was repeatedly formed and destroyed due to the severe pulverization and fracture caused by the huge periodic volume fluctuation during lithiation/delithiation, so that almost no protection existed for the active Si. Figure 6a and b schematically illustrates this process. Thus, the effectively suppressed pulverization warranted not only the integrity of the anode material but also ensures a stable SEI layer, which, we believe, is the most critical factor responsible for the significantly improved long-term cycling stability of the samples prepared from Si-$x$$LiBH_4$.

Figure 7 displays optical photographs and corresponding SEM images of the wafer electrodes after different cycles. As shown in the insets of Figure 7a-c, the exposed Cu current collector was clearly observed in the wafer electrode made of pristine Si due to the exfoliation of active materials only after 5 cycles. SEM observation confirmed the severe fracture and destruction of electrode surface. On the contrary, the wafer electrode made of the multicomponent layer-protected micron-Si composite displayed such a good integrity that nearly no exfoliation of active materials was observed, as shown in the insets of Figure 7d-f. Even after 200 cycles, only sparse cracks appeared on the electrode surface as observed by SEM (Figure 7f). As a result, a good long-term cyclability was successfully achieved by the micron-Si composite anode with a protecting $SiO_x$/C layer consisting of multicomponent nanocrystals and B species.

## CONCLUSIONS

In this work, the B-containing multicomponent conformal layer-protected Si composite microparticles were successfully synthesized by heating the Si-$x$LiBH$_4$ mixtures followed by a ball milling treatment under $CO_2$ atmosphere. The resulted Si composites were mainly composed of crystalline Si as core and a 30-50 nm thick amorphous $SiO_x$/C layer as shell that involves well-dispersed ultrafine nanocrystals of SiC and $Li_2SiO_3$ and amorphous species (B and $B_2O_3$). The ultrafine nanocrystals and B species in the protecting outer layer effectively improved the mechanical properties of amorphous $SiO_x$/C layer, which restricted greatly the volume expansion upon lithiation and at the same time stabilized the formed SEI layer, and finally led synergistically to a remarkable long-term cyclability for the micro-Si electrodes. The available specific capacity of the micro-Si anode with high-strength B-containing multicomponent $SiO_x$/C layer still stayed at 975 mAh $g^{-1}$ at 100 mA $g^{-1}$ even after 600 cycles, which is 10-fold of commercial graphite anode.

### ASSOCIATED CONTENT

**Supporting Information:** The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/

XRD and XPS spectra of post-heated Si-$x$LiBH$_4$, nanoindentation results of micron-Si composites with/without B, typical nanoindentation morphology of Si-0.3LiBH$_4$, particle size distribution of Si-$x$LiBH4, and electrochemical properties of Si-0.3LiBH$_4$ with/without prelithiation. (PDF)

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## NOTES

The authors declare no competing financial interest.

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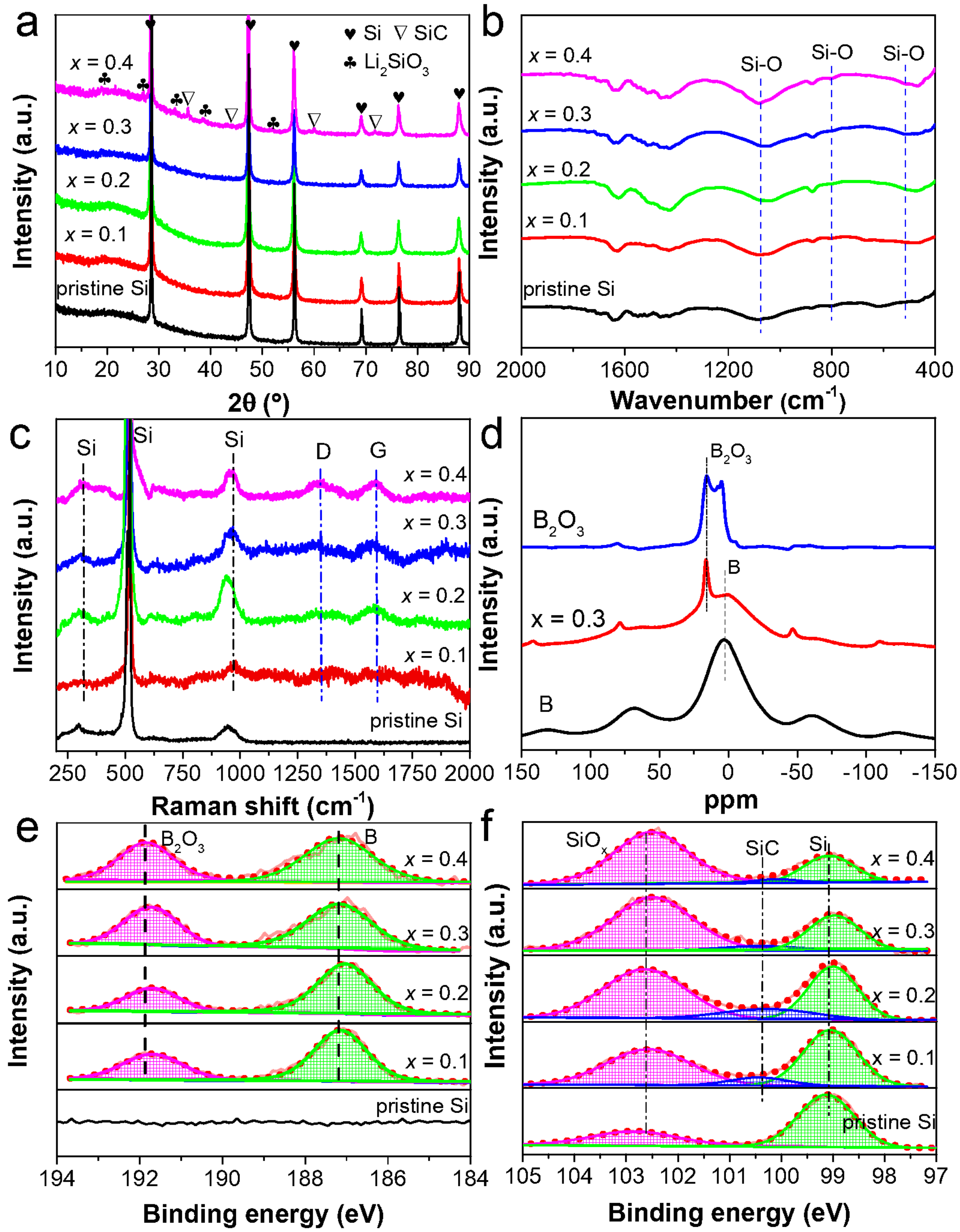



**Figure 1** (a) XRD, (b) FTIR, (c) Raman, (d) B11 NMR (e) B 1s and (f) Si 2p XPS spectra of the micron-Si composites prepared from Si-$x$LiBH$_4$ after 4h of milling.

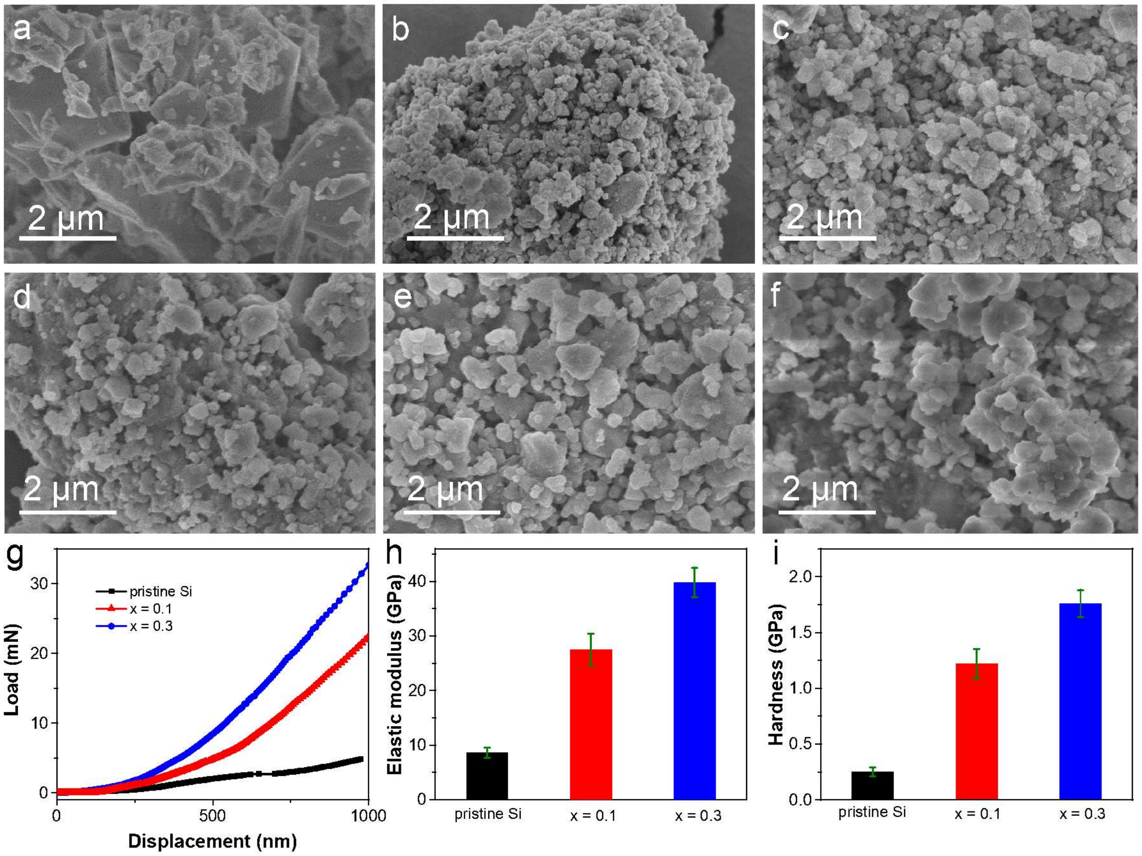



**Figure 2** SEM images of (a) pristine and (b) milled Si, and the micron-Si composites prepared from Si-$x$LiBH$_4$ (c) $x = 0.1$, (d) $x = 0.2$, (e) $x = 0.3$, and (f) $x = 0.4$. (g) Nanoindentation load/displacement curves, (h) elastic moduli and (i) hardness values measured for pristine Si and the prepared micron-Si composites.

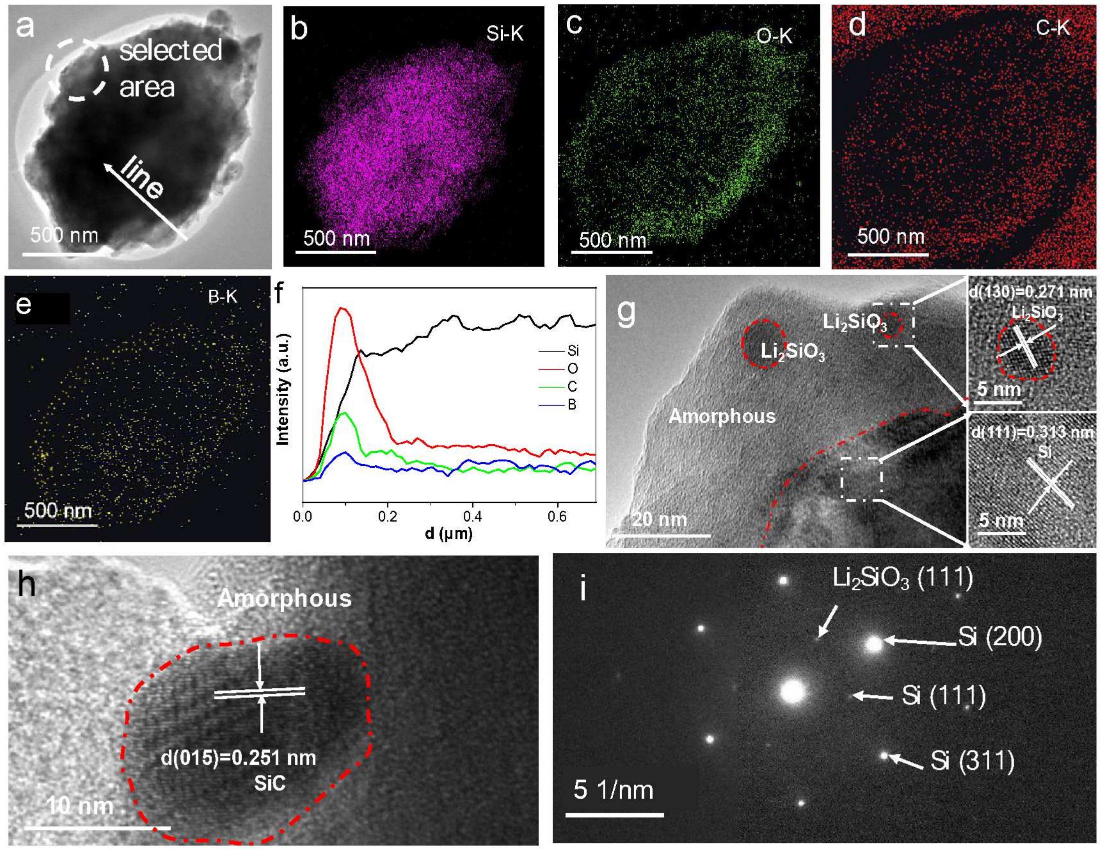



**Figure 3** (a) TEM image and the corresponding EDS mappings of (b) Si, (c) O, (d) C and (e) B distribution, (f) linear scanning, (g, h) HRTEM images, and (i) SAED pattern of the micron-Si composite prepared from Si-0.3$LiBH_4$ after 4h of milling.

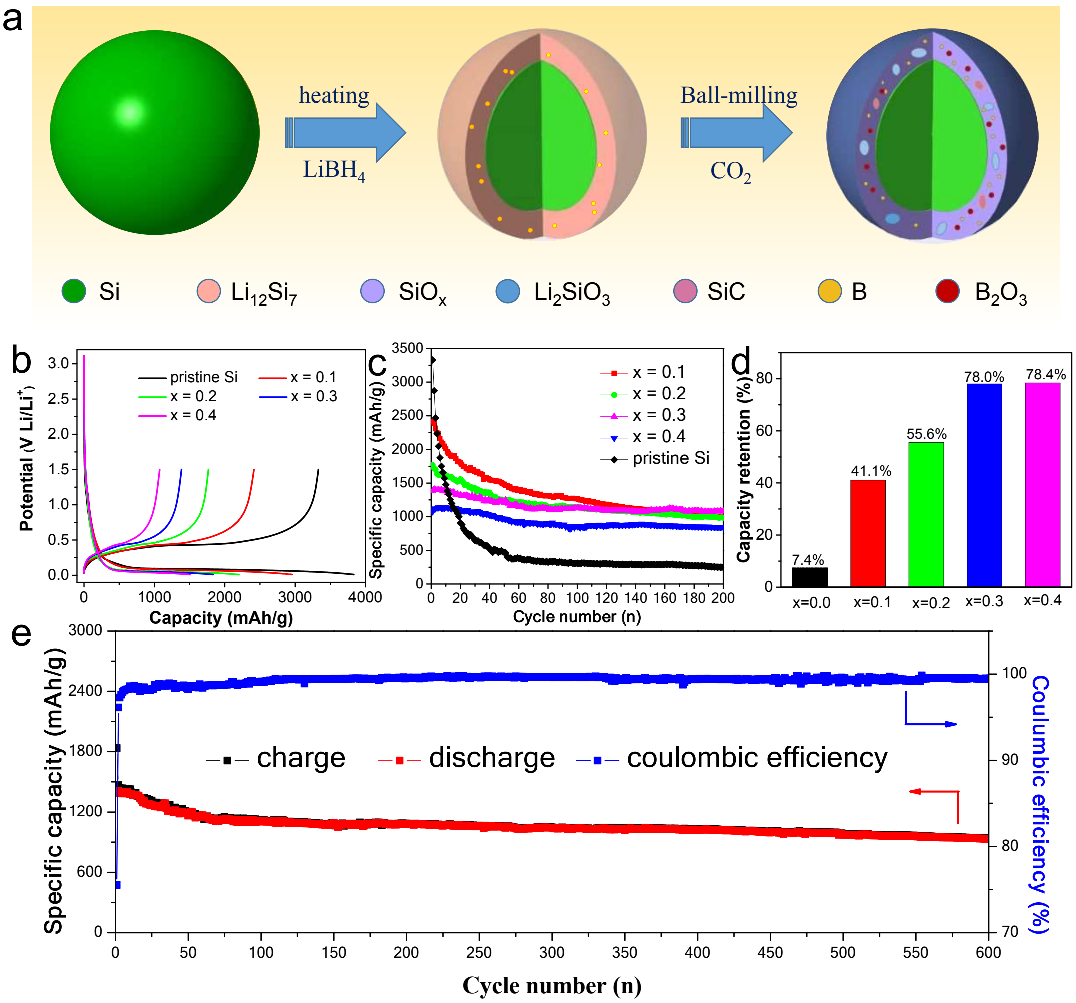



**Figure 4** (a) Illustration of fabrication procedure of the multicomponent layer-protected micron-Si composite. (b) The first voltage profiles, (c) cycling performance and (d) capacity retention of pristine Si and the multicomponent layer-protected micron-Si at 100 mA $g^{-1}$ and 0.01-1.5 V. (e) Specific capacity and coulombic efficiency of the micron-Si composite prepared from Si-0.3$LiBH_4$ within 600 cycles.

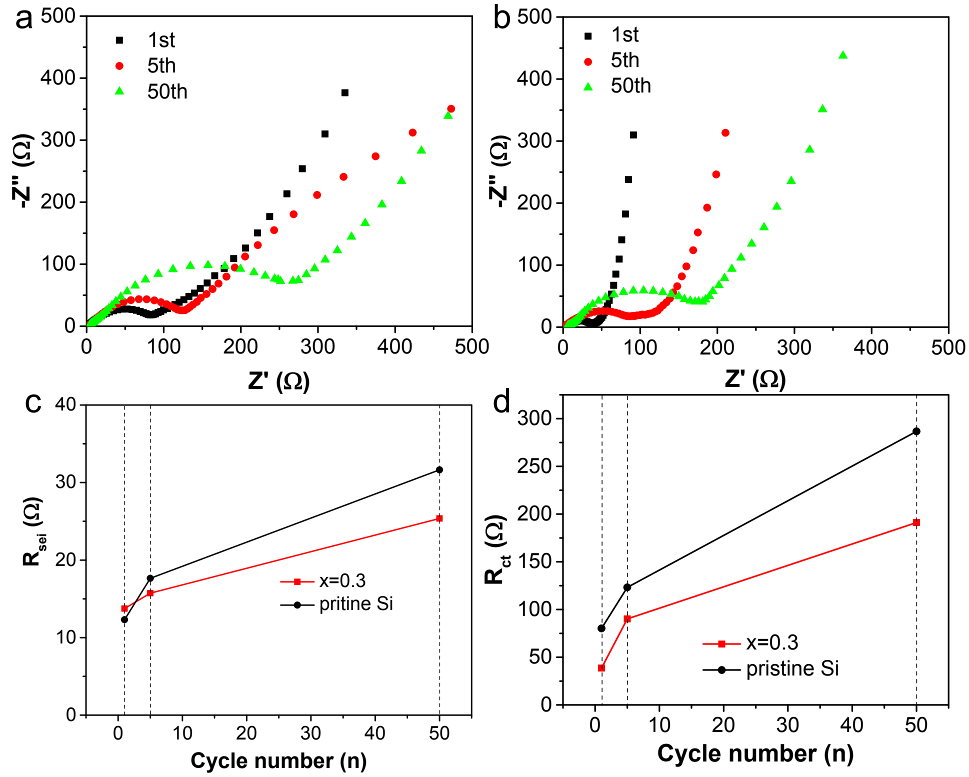



**Figure 5** EIS spectra of (a) pristine Si and (b) the micron-Si composite prepared from Si-0.3$LiBH_4$ as a function of cycles and the fitting results of (c) $R_{sei}$ and (d) $R_{ct}$.

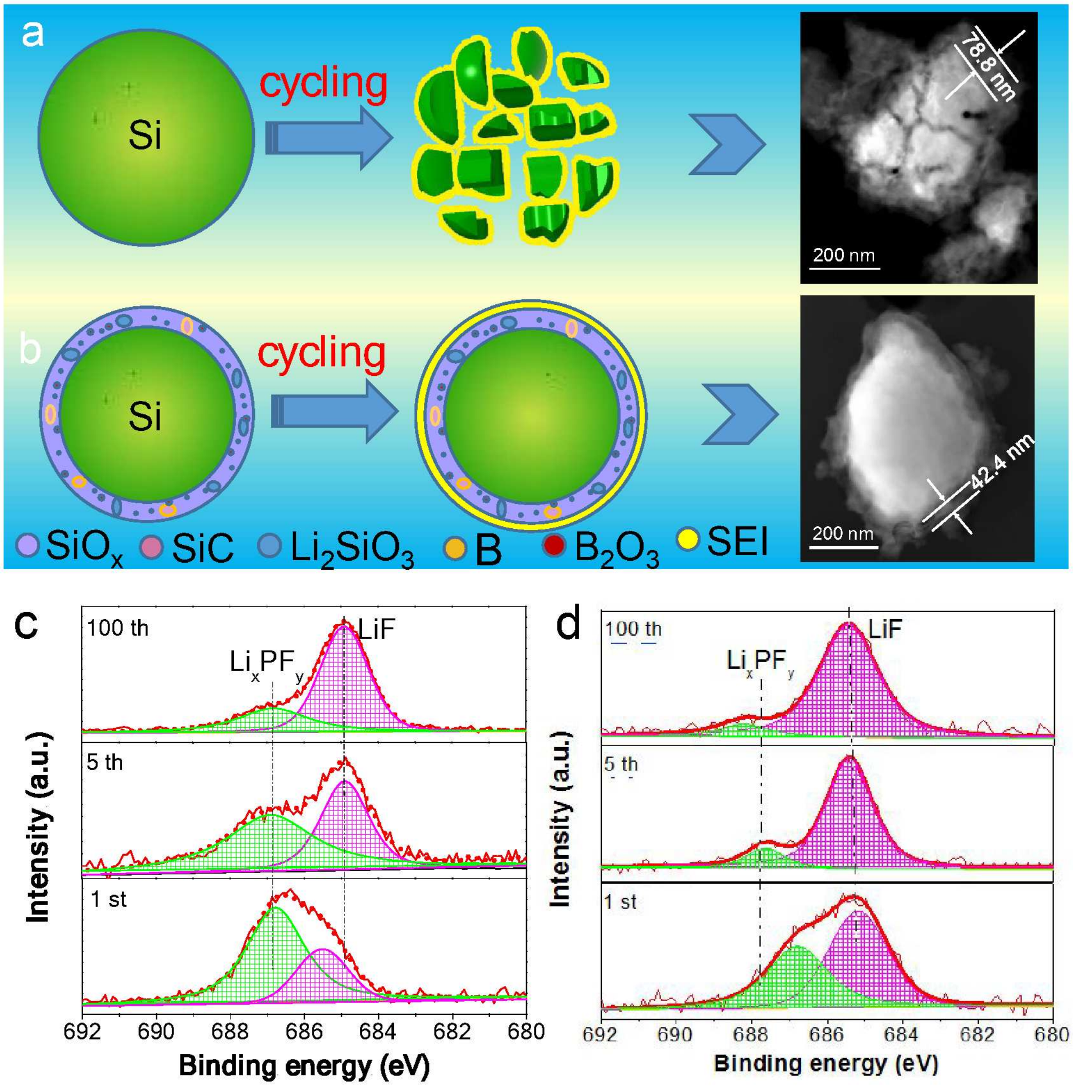



**Figure 6** Graphical illustration of volume expansion and pulverization of (a) Si microparticle and (b) the multicomponent layer-protected micron-Si composite during lithiation/delithiation cycling (STEM images are shown at the right side after 200 cycles). High-resolution XPS spectra of F 1s of (c) pristine Si and (d) the micron-Si composite prepared from Si-0.3$LiBH_4$ at selected cycles.

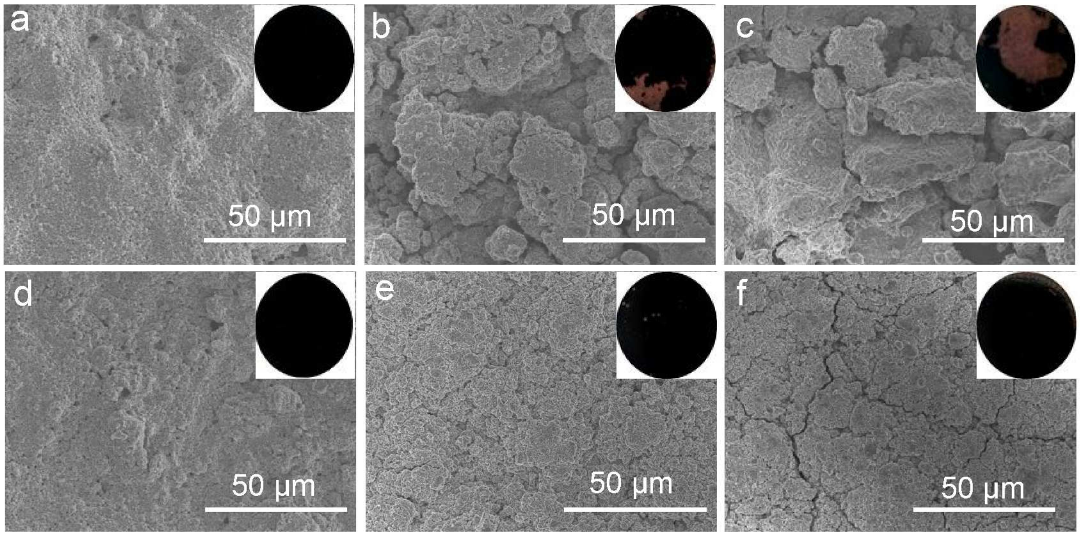
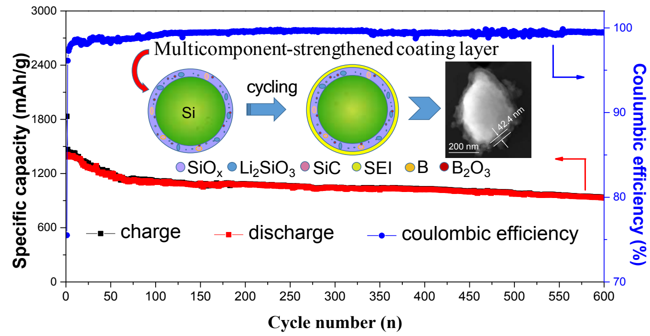



**Figure 7** SEM images of the electrode surface for pristine Si (a–c) and the micron-Si composite prepared from Si-0.3$LiBH_4$ (d–f). (a, d) after 1 cycle, (b, e) after 5 cycles, (c, f) after 200 cycles.

## Table of Contents Image




An in situ formed high-strengthened B-containing multicomponent coating layer enables remarkable long-term cyclability of micron-Si composite anode.